

1. What are the differences between Native, Hybrid, and Cross-platform mobile application development, and when is each approach commonly used?

Native, Hybrid, and Cross-platform are three different ways of developing mobile applications.

Native development creates apps for one operating system only, such as Android or iOS. It offers the best performance and user experience, making it ideal for apps like games, banking, and social media.

Hybrid development uses web technologies like HTML, CSS, and JavaScript to create one app that runs on different platforms. It is faster and less expensive to build but may not perform as well as native apps. It is commonly used for simple business apps or prototypes.

Cross-platform development uses frameworks like Flutter to build one application for both Android and iOS. It provides good performance while saving time and development costs, making it popular for many modern mobile applications.

2. What is the Flutter Ecosystem, and how does it help developers create modern mobile applications?

The Flutter Ecosystem is a set of tools, libraries, and resources created by Google for building mobile applications. It uses the Dart programming language and allows developers to create apps for Android and iOS using a single codebase.

Flutter includes ready-made widgets, Hot Reload for faster testing, and packages from **pub.dev** that add extra features. These tools help developers build attractive, fast, and reliable applications while reducing development time.

3. What is the Software Development Life Cycle (SDLC), and why is it important in building successful applications?

The Software Development Life Cycle (SDLC) is a step-by-step process used to develop software. It includes planning, designing, developing, testing, deploying, and maintaining an application.

SDLC is important because it helps developers stay organized, reduce errors, and make sure the application meets user needs. Following the SDLC also improves software quality and makes projects easier to manage.

4. How do Flutter development practices and the SDLC work together in creating real-world mobile applications?

Flutter supports every stage of the SDLC. During planning and design, developers decide how the app will work. During development, Flutter's Hot Reload speeds up coding and testing. After testing, the same app can be deployed to Android and iOS using one codebase. This makes development faster and maintenance easier.

5. Why should developers understand these concepts before starting an application project?

Developers should understand these concepts because they help in choosing the right development approach and planning the project properly. Knowing Flutter and the SDLC also helps save time, reduce mistakes, and create applications that are easier to maintain. A good understanding of these topics increases the chances of building a successful mobile application.

